

## Research paper

## Causal relationships between dietary habits and five major mental disorders: A two-sample Mendelian randomization study

Zhiqiang Du<sup>a,1</sup>, Shuaiyi Guo<sup>a,1</sup>, Yifan Sun<sup>a,1</sup>, Qin Zhou<sup>a</sup>, Ying Jiang<sup>a</sup>, Yuan Shen<sup>a</sup>, Haohao Zhu<sup>a,\*</sup>, Zhenhe Zhou<sup>a,\*</sup>, Hongliang Zhou<sup>b,\*</sup><sup>a</sup> Wuxi Mental Health Center, Nanjing Medical University, Wuxi, Jiangsu, China<sup>b</sup> Department of Psychology, The Affiliated Hospital of Jiangnan University, Wuxi, Jiangsu, China

## ARTICLE INFO

## Keywords:

Dietary habits  
Mental disorders  
Mendelian randomization  
Genome-Wide Association Study  
Gluten-free diet

## ABSTRACT

**Objective:** To explore the causal relationship between dietary habits and five major mental disorders using the two-sample Mendelian randomization (MR) analysis.**Methods:** This study was based on the summary data of the genome-wide association study (GWAS) on diet and five major mental disorders in the European population. The genetic locus data of five major mental disorders (mania, bipolar disorder, manic depression, depression, schizophrenia) from those who never eat eggs, dairy, wheat, and sugar were used. Two-sample MR analysis was conducted to evaluate the causal relationship between diet and five major mental disorders.**Results:** This study revealed a causal relationship between “Never eat Wheat products” and all five types of mental disorders (mania, bipolar disorder, manic depression, depression, schizophrenia), demonstrating a significant negative correlation ( $P < 0.05$ ). However, no significant causal relationship was observed between “Never eat Sugar or foods/drinks containing sugar” and any of the five mental disorders ( $P > 0.05$ ). Furthermore, the study found that the statement “Never eat eggs, dairy, wheat, sugar: I eat all of the above” had a causal relationship with mania, bipolar disorder, and manic depression, showing a significant positive correlation ( $P < 0.05$ ). However, this statement did not exhibit a significant causal relationship with depression and schizophrenia ( $P > 0.05$ ).**Conclusion:** There was a negative correlation between never eating wheat products and the five mental disorders (mania, bipolar disorder, manic depression, depression, schizophrenia), indicating that never eating wheat products may reduce the risk of mental disorders.

## 1. Introduction

Mental disorders refer to the disruption of brain function due to biological, psychological, and social factors, leading to cognitive, emotional, volitional, and behavioral impairments in individuals (Huang et al., 2019a; Sagar et al., 2020; Liu et al., 2017). Currently, mental disorders have become a major public health issue worldwide, with a global lifetime prevalence rate of adult mental disorders ranging from 12.2 % to 48.6 % (Karyotaki et al., 2020). Additionally, mental disorders are strong risk factors for suicide; approximately 70 % of suicides are caused by mental disorders (Gili et al., 2019; Wu et al., 2020; Cooper, 2019). Therefore, it is urgent to identify potential causal risk factors for mental disorders.

Although the etiology of mental disorder is not entirely clear, studies have shown that most mental disorders are related to diet, and diet may be crucial for the onset and treatment of mental disorders (Ljungberg et al., 2020). Schizophrenia is a severe mental disorder characterized by deficits in thinking and emotions, hallucinations, delusions, and impaired social functioning. It imposes significant psychological, social, and economic burdens on patients and their families, with substantial impacts on societal order and public health. Research has found that individuals with schizophrenia tend to consume higher amounts of sugar and saturated fats in their diets, while their fiber intake is reduced, although study results vary (Firth et al., 2018; Aucoin et al., 2020; Bly et al., 2014; Teasdale et al., 2019). Bipolar disorder, also known as manic depression, is a common mental disorder characterized by intense

\* Corresponding authors.

E-mail addresses: [zhuhh@jiangnan.edu.cn](mailto:zhuhh@jiangnan.edu.cn) (H. Zhu), [zhouzh@njmu.edu.cn](mailto:zhouzh@njmu.edu.cn) (Z. Zhou), [Hongliangzh2022@hotmail.com](mailto:Hongliangzh2022@hotmail.com) (H. Zhou).<sup>1</sup> These authors contributed equally to this work.

mood swings, including episodes of depression and episodes of elevated mood. This condition profoundly affects patients' social functioning, mental health, and overall quality of life. Studies on dietary assessment for bipolar disorder are relatively limited, but dietary and nutritional intake patterns are generally comparable to those observed in schizophrenia patients (Bly et al., 2014; Teasdale et al., 2019). Mania is a prevalent mental disorder characterized by symptoms such as elevated mood, excessive energy, impulsive behavior, increased speech, and lack of concentration. Mania significantly impacts patients' daily lives, work, and social functioning, and it is associated with an increased risk of suicide. Research has indicated that lower dietary protein intake is related to cognitive decline in recent cases of mania (Dickerson et al., 2020). Depression, a common psychological disorder, features symptoms like low mood, loss of interest and pleasure, lack of energy, difficulty concentrating, decreased self-esteem, sleep disturbances, and changes in appetite. Studies have suggested a connection between diets rich in added sugars, sugary beverages, and junk food and an increased risk of depression (Huang et al., 2019b; Lassale et al., 2019; Nicolaou et al., 2020).

Dietary factors are believed to be related to inflammation and various health outcomes, including mental health and cognitive function (Hepsomali and Coxon, 2022). Excessive eating or lack of dietary mindfulness can potentially lead to adverse psychological conditions such as mood depression and emotional instability (Nightingale and Cassin, 2019). Therefore, cultivating healthy eating habits contributes to maintaining positive mental and emotional well-being. Adopting healthy dietary habits can foster a proper mindset towards food by encouraging the selection of nutritious options. Under the guidance of a well-balanced nutritional diet, these habits can further facilitate the development of individuals' mental and emotional health. However, there has been basically no research at home and abroad to determine whether there is a causal relationship between diet and mental disorders. Conducting research on the causal relationship between diet and five types of mental disorders serves two significant purposes. On one hand, it helps to address the existing gaps in this field of study. On the other hand, it offers insights into the causal links between diet and these disorders, which can be beneficial for clinical interventions. The intriguing aspect of these research findings lies in the modifiability of diet, presenting opportunities for dietary interventions. Such interventions are comparatively cost-effective and pose lower risks compared to treatments involving psychology or medication. Moreover, patients are more likely to be receptive to dietary interventions.

In recent years, Mendelian randomization (MR) research has been widely applied. This technique, a type of instrumental variable analysis, effectively uses existing Genome-Wide Association Study (GWAS) data results, with genetic variants as instrumental variables (IVs), to explore the causal relationship between exposure factors and outcomes (Hemani et al., 2018; Woolf et al., 2022; Lv et al., 2023). MR research is based on Mendel's second law of genetics, grouping the study population based on whether they carry a certain genetic variation, and then comparing the occurrence of outcomes between the two groups. This grouping method will not be interfered with by other characteristics, environment, and socioeconomic status of the population; at the same time, because the formation of the exposure part explained by genetic variation precedes the outcome, the interference of reverse causality can also be excluded, effectively making up for the shortcomings of traditional methods. Therefore, in this study, we use MR based on two-sample summary data to analyze the causal relationship between the diet: "Never eat eggs, dairy, wheat, sugar" and "I eat all of the above", and the five major mental disorders: mania, bipolar disorder, manic depression, depression, and schizophrenia. This provides new ideas for the clinical prevention and treatment of mental disorders.

## 2. Methods

### 2.1. Source of sample data

The genetic data of the exposure phenotype in this study were all sourced from the Medical Research Council-Integrative Epidemiology Unit (MRC-IEU), specifically from European populations. The summary statistical data on diet (including 'never eat eggs', 'dairy', 'wheat', 'sugar', and 'I eat all of the above') were obtained using the output from the GWAS pipeline, which utilized Pheasant derived variables from the UKBiobank (Table 1).

The genetic data for the outcome phenotype in this study were derived from large-scale GWAS-meta studies, again from European populations. Data on non-cancer illness codes, self-reported for mania/bipolar disorder/manic depression and depression, were sourced from the Medical Research Council-Integrative Epidemiology Unit (MRC-IEU), utilizing output from the GWAS pipeline using Pheasant derived variables from the UKBiobank. Data on schizophrenia were sourced from the Schizophrenia Working Group of the Psychiatric Genomics Consortium (Table 2). All genetic data could be downloaded from the MR BASE database at <http://app.mrbase.org/>.

### 2.2. Instrumental variable selection

The selection criteria for single nucleotide polymorphisms (SNPs) related to exposure from GWAS summary data were as follows: 1) achieving genome-wide research significance,  $P < 5 \times 10^{-8}$ ; 2) excluding SNPs in linkage disequilibrium to ensure SNPs are mutually independent ( $r^2 < 0.01$ , window size = 5000 KB); 3) corresponding relevant data in the summary data of the outcome GWAS. In this study, 'Never eat eggs', 'dairy', 'wheat', 'sugar', and 'I eat all of the above' were considered as exposure phenotypes, and the five major mental disorders: mania, bipolar disorder, manic depression, depression, and schizophrenia were considered as outcomes, to screen for relevant SNPs. Ultimately, no relevant SNPs were screened in the exposure factors: 'Never eat eggs', 'dairy' with the outcome: five major mental disorders. However, relevant SNPs were screened in 'Never eat wheat', 'sugar' and 'I eat all of the above' with the five mental disorders (Table 3).

The corresponding data of the instrumental variables (IV) in the exposure and outcome were combined, including chromosomal location information, effect allele, other alleles, effect allele frequency, effect estimate for the phenotype, standard error of the effect estimate, and  $P$ -value from both exposure and outcome GWAS summary data. We excluded IVs that couldn't be corrected for forward and reverse strands based on MAF due to strand confusion. We used MR-PRESSO (Mendelian randomization pleiotropy residual sum and outlier) to remove IVs with

**Table 1**  
Summary of genetic data information for exposure phenotypes.

| Phenotypes                                                                  | GWAS ID      | Sample size | SNP       | Pubmed ID (or URL)                                          |
|-----------------------------------------------------------------------------|--------------|-------------|-----------|-------------------------------------------------------------|
| Never eat eggs, dairy, wheat, sugar: Wheat products                         | ukb-b-3599   | 461,046     | 9,851,867 | <a href="http://app.mrbase.org/">http://app.mrbase.org/</a> |
| Never eat eggs, dairy, wheat, sugar: I eat all of the above                 | ukb-b-2393   | 461,046     | 9,851,867 | <a href="http://app.mrbase.org/">http://app.mrbase.org/</a> |
| Never eat eggs, dairy, wheat, sugar: Dairy products                         | ukb-b-18,909 | 461,046     | 9,851,867 | <a href="http://app.mrbase.org/">http://app.mrbase.org/</a> |
| Never eat eggs, dairy, wheat, sugar: Sugar or foods/drinks containing sugar | ukb-b-5495   | 461,046     | 9,851,867 | <a href="http://app.mrbase.org/">http://app.mrbase.org/</a> |
| Never eat eggs, dairy, wheat, sugar: Eggs or foods containing eggs          | ukb-b-17,455 | 461,046     | 9,851,867 | <a href="http://app.mrbase.org/">http://app.mrbase.org/</a> |

**Table 2**  
Summary of genetic data information for outcome phenotypes.

| Phenotypes                              | GWAS ID      | Sample size | SNP        | Pubmed ID(or URL)                                           |
|-----------------------------------------|--------------|-------------|------------|-------------------------------------------------------------|
| Schizophrenia                           | ieu-b-42     | 77,096      | 15,358,497 | 25056061                                                    |
| Mania/bipolar disorder/manic depression | ukb-b-6906   | 462,933     | 9,851,867  | <a href="http://app.mrbase.org/">http://app.mrbase.org/</a> |
| Depression                              | ukb-b-12,064 | 462,933     | 9,851,867  | <a href="http://app.mrbase.org/">http://app.mrbase.org/</a> |

**Table 3**  
Results of diet and mental disorder screening SNPs.

| Phenotypes |                                        | Outcome       |                                         |            |
|------------|----------------------------------------|---------------|-----------------------------------------|------------|
|            |                                        | Schizophrenia | Mania/bipolar disorder/manic depression | Depression |
| Exposure   | Never eat eggs, dairy, wheat, sugar:   | +             | +                                       | +          |
|            | Wheat products                         |               |                                         |            |
|            | Never eat eggs, dairy, wheat, sugar:   | +             | +                                       | +          |
|            | I eat all of the above                 |               |                                         |            |
|            | Never eat eggs, dairy, wheat, sugar:   | –             | –                                       | –          |
|            | Dairy products                         |               |                                         |            |
|            | Never eat eggs, dairy, wheat, sugar:   | +             | +                                       | +          |
|            | Sugar or foods/drinks containing sugar |               |                                         |            |
|            | Never eat eggs, dairy, wheat, sugar:   | –             | –                                       | –          |
|            | Eggs or foods containing eggs          |               |                                         |            |

The “+” represents the selected relevant SNPs, while the “–” indicates that no relevant SNPs were selected.

horizontal pleiotropy. By removing IVs one by one and re-performing MR analysis, we could identify which IVs made the results more significant when removed, and the difference between the results after removal and the original results was significant (Verbanck et al., 2018).

We evaluated whether the chosen IVs had weak IV bias by calculating the F statistic. If  $F > 10$ , it indicates that there is no weak IV bias, further validating the association hypothesis. The formula for calculating F is  $[(N-K-1)/K] \times [R^2/(1-R^2)]$ , where N is the sample size of the exposure factor, K is the number of IVs, and R<sup>2</sup> is the proportion of exposure factor variance explained by the IVs.

2.3. MR analysis

Inverse variance weighted (IVW), weighted median, weighted mode, and MR Egger were applied for two-sample MR analysis. IVW provides unbiased results under the validity assumption (no horizontal pleiotropy of IV), but the presence of pleiotropy would affect the results. Weighted median and weighted mode estimates remain robust even if a small number of IVs do not meet the validity assumption, but they are not unbiased estimates. MR Egger introduces an intercept term, considers pleiotropy, but its estimate result has low power. In this study, the results obtained by IVW are used as the main research results, and the causal effects estimated by other methods are used as sensitivity analysis results. Meanwhile, to better interpret the results, the Beta values obtained in the study were converted to odds ratios (OR), and the 95 % confidence intervals (CI) were calculated.

The study used MR Egger to test for horizontal pleiotropy, which assumes no such pleiotropy, i.e., IVs affect the outcome entirely through the exposure. If the results show  $P > 0.05$ , it indicates that there is no

statistically significant difference between the intercept and 0, implying no issue of horizontal pleiotropy. The data were subjected to “leave-one-out” analysis, sequentially excluding one SNP and calculating the Mendelian randomization effect of the remaining SNPs. By displaying through a forest plot, the influence of each SNP on the results can be intuitively judged, thus determining the stability of the TSMR analysis results. All MR analyses were performed on the MR Base platform (App version: 1.2.1 e646be [27 June 2018], R version: 3.5.0) (Hemani et al., 2016).

3. Results

3.1. Causal relationship between never eat wheat products and mental disorders

In the study of the causal relationship between never eating wheat products and mental disorder, MR analysis was conducted with never eating wheat products as the exposure factor and five mental disorders as the outcome. In the analysis of the causal relationship between never eating wheat products and mania/bipolar disorder/manic depression, 18 IVs related to never eating wheat products were initially selected from the GWAS database. After removing 10 abnormal IVs with linkage disequilibrium, 8 IVs with genome-wide significant associations were ultimately selected. Horizontal pleiotropy was checked with MR Egger and MR-PRESSO and was deemed non-existent ( $P$  for MR-Egger intercept =  $0.489 > 0.05$ ). The IVW method results showed a causal relationship between never eating wheat products and mania/bipolar disorder/manic depression, exhibiting a negative correlation ( $OR: 0.953$ ,  $P = 9.39E-09 < 0.05$ ). The other three methods further confirmed a causal relationship between never eating wheat products and mania/bipolar disorder/manic depression, showing a negative correlation ( $P < 0.05$ ) (Table 4, Fig. 1A).

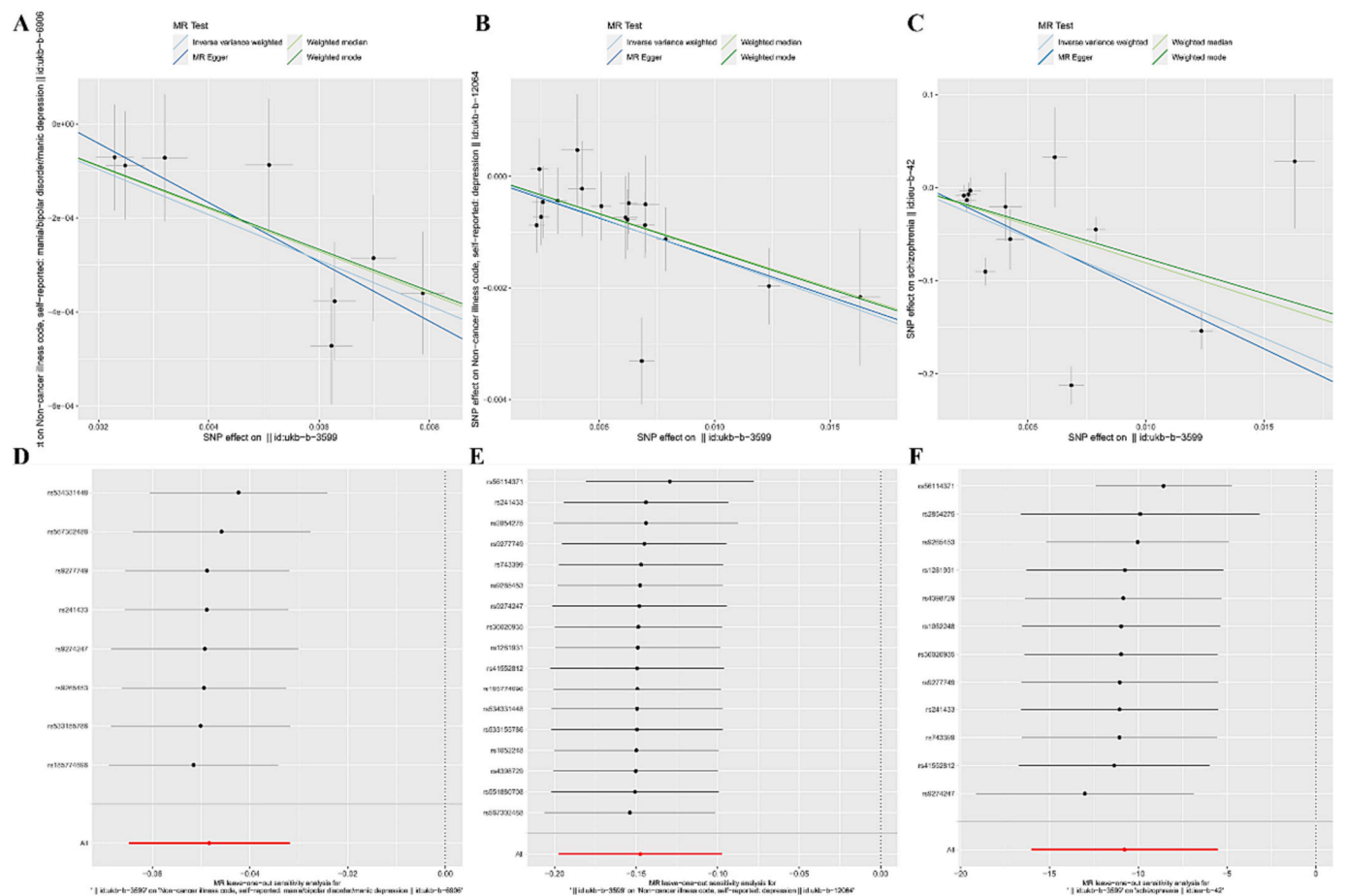
In the analysis of the causal relationship between never eating wheat products and depression, 18 IVs related to never eating wheat products were initially selected from the GWAS database. After removing one abnormal IV with linkage disequilibrium, 17 IVs with genome-wide significant associations were ultimately selected. Horizontal pleiotropy was checked with MR Egger and MR-PRESSO and was deemed non-existent ( $P$  for MR-Egger intercept =  $0.892 > 0.05$ ). The IVW method results showed a causal relationship between never eating wheat products and depression, exhibiting a negative correlation ( $OR: 0.863$ ,  $P = 8.50E-09 < 0.05$ ). The other three methods further confirmed a causal relationship between never eating wheat products and depression, showing a negative correlation ( $P < 0.05$ ) (Table 4, Fig. 1B).

In the analysis of the causal relationship between never eating wheat products and schizophrenia, 18 IVs related to never eating wheat products were initially selected from the GWAS database. After removing 6 abnormal IVs with linkage disequilibrium, 12 IVs with genome-wide significant associations were ultimately selected. Horizontal pleiotropy was checked with MR Egger and MR-PRESSO and was deemed non-existent ( $P$  for MR-Egger intercept =  $0.747 > 0.05$ ). The IVW method results showed a causal relationship between never eating wheat products and schizophrenia, exhibiting a negative correlation ( $OR: 2.08E-05$ ,  $P = 5.88E-05 < 0.05$ ). The other three methods further confirmed a causal relationship between never eating wheat products and schizophrenia, showing a negative correlation ( $P < 0.05$ ) (Table 4, Fig. 1C).

In the MR analysis of the causal relationship between never eating wheat products and the five mental disorders, all tool variable F values are  $> 10$ , indicating that there is no weak offset in all results, confirming that the results are reliable. The heterogeneity result  $P > 0.05$ , indicates that there is no heterogeneity in the included IVs, and the impact of heterogeneity does not need to be considered in the results. Also, a “leave-one-out” analysis was performed on all data, and after testing by individual removal, no single SNP has a large effect on the robustness of the results, hence this study is stable (Fig. 1D, E, F). Therefore, the MR

**Table 4**  
MR analysis of the causal relationship between never eating wheat products and mental disorders.

| Outcome Phenotypes                      | MR method                 | Number of SNPs | Beta    | SE    | OR(95 % confidence Interval) | Association P-value | P for MR-Egger intercept |
|-----------------------------------------|---------------------------|----------------|---------|-------|------------------------------|---------------------|--------------------------|
| Mania/bipolar disorder/manic depression | MR Egger                  | 8              | −0.063  | 0.022 | 0.939(0.900–0.980)           | 0.027               | 0.489                    |
|                                         | Weighted median           | 8              | −0.045  | 0.011 | 0.956(0.936–0.976)           | 3.50E-05            |                          |
|                                         | Inverse variance weighted | 8              | −0.048  | 0.008 | 0.953(0.937–0.969)           | 9.39E-09            |                          |
|                                         | Weighted mode             | 8              | −0.045  | 0.014 | 0.956(0.931–0.982)           | 0.014               |                          |
| Depression                              | MR Egger                  | 17             | −0.141  | 0.053 | 0.869(0.783–0.963)           | 0.017               | 0.892                    |
|                                         | Weighted median           | 17             | −0.133  | 0.033 | 0.876(0.821–0.934)           | 5.19E-05            |                          |
|                                         | Inverse variance weighted | 17             | −0.147  | 0.026 | 0.863(0.821–0.907)           | 8.50E-09            |                          |
|                                         | Weighted mode             | 17             | −0.135  | 0.036 | 0.874(0.814–0.938)           | 0.002               |                          |
| Schizophrenia                           | MR Egger                  | 12             | −12.100 | 4.861 | 5.56E-06(4.05E-10-0.076)     | 0.032               | 0.747                    |
|                                         | Weighted median           | 12             | −8.093  | 1.834 | 0.0003(8.4E-06-0.011)        | 1.02E-05            |                          |
|                                         | Inverse variance weighted | 12             | −10.780 | 2.682 | 2.08E-05(1.08E-07-0.004)     | 5.88E-05            |                          |
|                                         | Weighted mode             | 12             | −7.579  | 1.560 | 0.001(2.4E-05-0.011)         | 0.001               |                          |



**Fig. 1.** Causal relationship between never eat wheat products and mental disorders. A: Scatter plot of the causal relationship between “Never eat Wheat products” and mania/bipolar disorder/manic depression. The slope of each line represents the causal relationship of each method. B: Scatter plot of the causal relationship between “Never eat Wheat products” and depression. The slope of each line represents the causal relationship of each method. C: Scatter plot of the causal relationship between “Never eat Wheat products” and schizophrenia. The slope of each line represents the causal relationship of each method. D: Leave-one-out plot of the causal relationship between “Never eat Wheat products” and mania/bipolar disorder/manic depression. E: Leave-one-out plot of the causal relationship between “Never eat Wheat products” and depression. F: Leave-one-out plot of the causal relationship between “Never eat Wheat products” and schizophrenia.

analysis results support a causal relationship between never eating wheat products and the five mental disorders, all showing a negative correlation.

### 3.2. The causal relationship between never eating sugar or foods/drinks containing sugar and mental disorder

In the study of the causal relationship between never eating sugar or foods/drinks containing sugar and mental disorder, we used never eating sugar or foods/drinks containing sugar as the exposure factor,



and five kinds of mental disorders as the outcome for MR analysis.

In the causal analysis between never eating sugar or foods/drinks containing sugar and mania/bipolar disorder/manic depression, 22 IVs related to never eating sugar or foods/drinks containing sugar were initially selected from the GWAS database. After removing 6 abnormal IVs with continuous imbalances, 16 IVs with genome-wide significance were finally selected. Simultaneously, horizontal pleiotropy was checked by MR Egger and MR-PRESSO, and it was thought that there was no horizontal pleiotropy ( $P$  for MR-Egger intercept = 0.309 > 0.05). The IVW method results show that there is no causal relationship between never eating sugar or foods/drinks containing sugar and mania/bipolar disorder/manic depression (OR:0.996,  $P$  > 0.05). The other three methods further prove that there is no causal relationship between never eating sugar or foods/drinks containing sugar and mania/bipolar disorder/manic depression ( $P$  > 0.05) (Table 5, Fig. 2A).

In the causal analysis between never eating sugar or foods/drinks containing sugar and depression, 22 IVs related to never eating sugar or foods/drinks containing sugar were initially selected from the GWAS database. After removing 2 abnormal IVs with continuous imbalances, 20 IVs with genome-wide significance were finally selected. Simultaneously, horizontal pleiotropy was checked by MR Egger and MR-PRESSO, and it was thought that there was no horizontal pleiotropy ( $P$  for MR-Egger intercept = 0.879 > 0.05). The IVW method results show that there is no causal relationship between never eating sugar or foods/drinks containing sugar and depression (OR:1.031,  $P$  > 0.05). The other three methods further prove that there is no causal relationship between never eating sugar or foods/drinks containing sugar and depression ( $P$  > 0.05) (Table 5, Fig. 2B).

In the causal analysis between never eating sugar or foods/drinks containing sugar and schizophrenia, 22 IVs related to never eating sugar or foods/drinks containing sugar were initially selected from the GWAS database. After removing 2 abnormal IVs with continuous imbalances, 20 IVs with genome-wide significance were finally selected. Simultaneously, horizontal pleiotropy was checked by MR Egger and MR-PRESSO, and it was thought that there was no horizontal pleiotropy ( $P$  for MR-Egger intercept = 0.12 > 0.05). The IVW method results show that there is no causal relationship between never eating sugar or foods/drinks containing sugar and schizophrenia (OR:3.688,  $P$  > 0.05). The other three methods further prove that there is no causal relationship between never eating sugar or foods/drinks containing sugar and schizophrenia ( $P$  > 0.05) (Table 5, Fig. 2C).

In the MR analysis of the causal relationship between never eating sugar or foods/drinks containing sugar and the five mental disorders, all tool variable  $F$  values are >10, indicating that there is no weak offset in all results, confirming that the results are reliable. The heterogeneity result  $P$  > 0.05, indicates that there is no heterogeneity in the included

IVs, and the impact of heterogeneity does not need to be considered in the results. Also, a “leave-one-out” analysis was performed on all data, and after testing by individual removal, no single SNP has a large effect on the robustness of the results, hence this study is stable (Fig. 2D, E, F). Therefore, the MR analysis results support that there is no causal relationship between never eating sugar or foods/drinks containing sugar and the five mental disorders.

### 3.3. Causal relationship between “never eat eggs, dairy, wheat, sugar: I eat all of the above” and mental disorder

In the research on the causal relationship between “Never eat eggs, dairy, wheat, sugar: I eat all of the above” and mental disorder, “Never eat eggs, dairy, wheat, sugar: I eat all of the above” was taken as the exposure factor, and five kinds of mental disorders as the outcome for MR analysis.

In the causal relationship analysis between “Never eat eggs, dairy, wheat, sugar: I eat all of the above” and mania/bipolar disorder/manic depression, 30 IVs related to “Never eat eggs, dairy, wheat, sugar: I eat all of the above” were initially selected from the GWAS database. After removing 11 abnormal IVs with continuous imbalance, 19 IVs with genome-wide significant correlation were finally screened out. The horizontal pleiotropy was checked by MR Egger and MR-PRESSO, and it was believed that there was no horizontal pleiotropy ( $P$  for MR-Egger intercept = 0.080 > 0.05). The result of IVW method showed that there was no causal relationship between “Never eat eggs, dairy, wheat, sugar: I eat all of the above” and mania/bipolar disorder/manic depression (OR:1.008,  $P$  > 0.05). But the other three methods all showed that “Never eat eggs, dairy, wheat, sugar: I eat all of the above” has a causal relationship with mania/bipolar disorder/manic depression, and there is a positive correlation ( $P$  < 0.05) (Table 6, Fig. 3A).

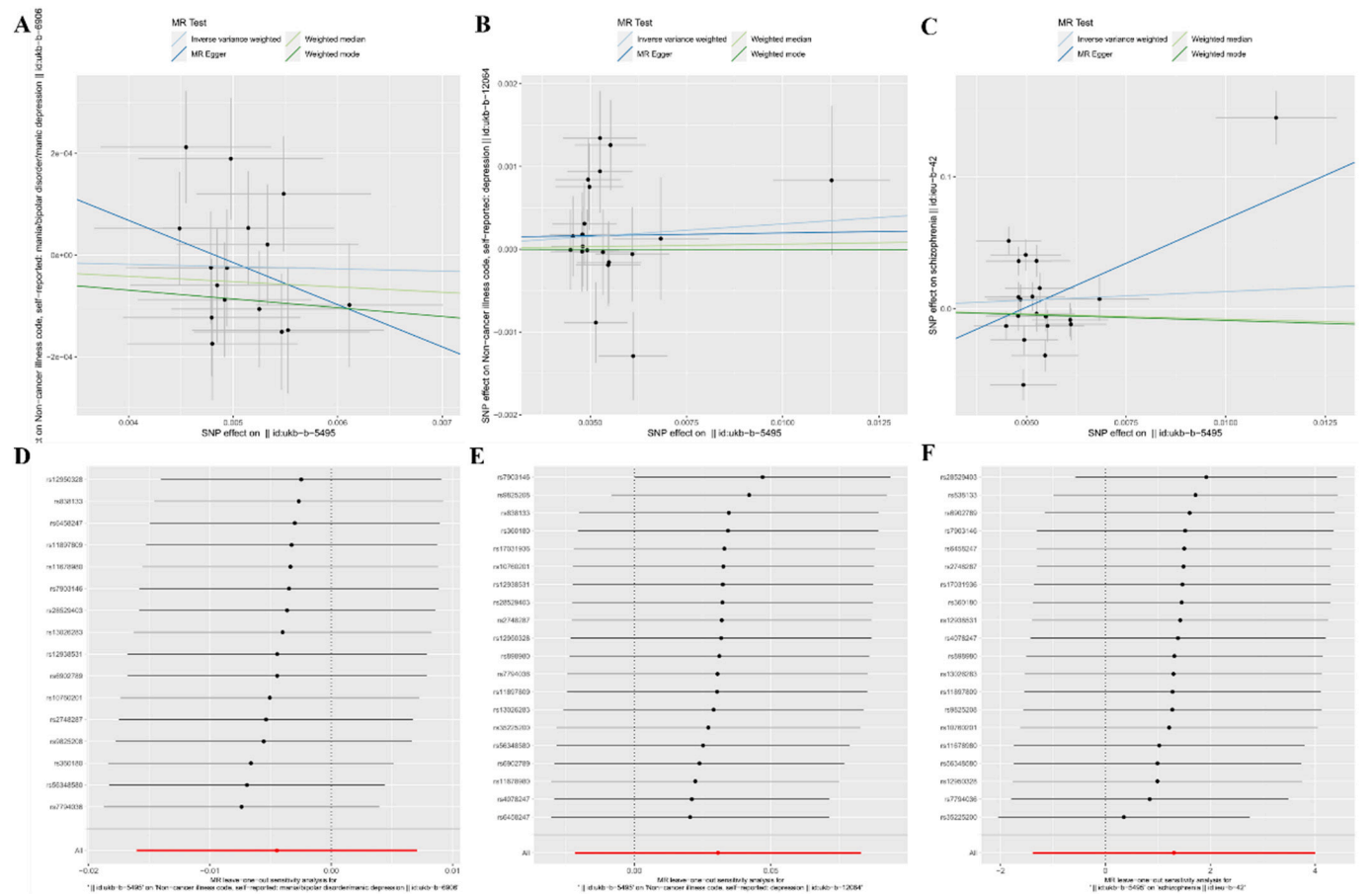
The causal relationship between “Never eat eggs, dairy, wheat, sugar: I eat all of the above” and depression was analyzed. 30 IVs related to “Never eat eggs, dairy, wheat, sugar: I eat all of the above” were initially selected from the GWAS database. After removing 1 abnormal IV with continuous imbalance, 29 IVs with genome-wide significant correlation were finally screened out. The horizontal pleiotropy was checked by MR Egger and MR-PRESSO, and it was believed that there was no horizontal pleiotropy ( $P$  for MR-Egger intercept = 0.336 > 0.05). The result of IVW method showed that there was no causal relationship between “Never eat eggs, dairy, wheat, sugar: I eat all of the above” and depression (OR:0.993,  $P$  > 0.05). The other three methods further proved that there was no causal relationship between “Never eat eggs, dairy, wheat, sugar: I eat all of the above” and depression ( $P$  > 0.05) (See Table 6, Fig. 3B).

In the causal relationship analysis between “Never eat eggs, dairy,

**Table 5**

MR analysis of the causal relationship between never eating sugar or foods/drinks containing sugar and mental disorders.

| Outcome Phenotypes                      | MR method                 | Number of SNPs | Beta   | SE    | OR(95 % confidence Interval) | Association P-value | P for MR-Egger intercept |
|-----------------------------------------|---------------------------|----------------|--------|-------|------------------------------|---------------------|--------------------------|
| Mania/bipolar disorder/manic depression | MR Egger                  | 16             | -0.083 | 0.075 | 0.920(0.795–1.065)           | 0.285               | 0.309                    |
|                                         | Weighted median           | 16             | -0.010 | 0.008 | 0.990(0.975–1.005)           | 0.176               |                          |
|                                         | Inverse variance weighted | 16             | -0.004 | 0.006 | 0.996(0.984–1.007)           | 0.446               |                          |
|                                         | Weighted mode             | 16             | -0.017 | 0.014 | 0.983(0.957–1.009)           | 0.224               |                          |
| Depression                              | MR Egger                  | 20             | 0.007  | 0.157 | 1.007(0.740–1.369)           | 0.966               | 0.879                    |
|                                         | Weighted median           | 20             | 0.006  | 0.032 | 1.006(0.946–1.070)           | 0.843               |                          |
|                                         | Inverse variance weighted | 20             | 0.031  | 0.027 | 1.031(0.978–1.087)           | 0.252               |                          |
|                                         | Weighted mode             | 20             | 0.000  | 0.057 | 1.000(0.894–1.118)           | 0.996               |                          |
| Schizophrenia                           | MR Egger                  | 20             | 13.320 | 7.474 | 609,259.8<br>(0.265–1.4E+12) | 0.092               | 0.12                     |
|                                         | Weighted median           | 20             | -0.773 | 0.870 | 0.462(0.084–2.541)           | 0.374               |                          |
|                                         | Inverse variance weighted | 20             | 1.305  | 1.374 | 3.688(0.250–54.491)          | 0.342               |                          |
|                                         | Weighted mode             | 20             | -0.881 | 1.028 | 0.414(0.055–3.106)           | 0.402               |                          |



**Fig. 2.** Causal relationship between Never Eating Sugar or Foods/Drinks Containing Sugar and Mental Disorder. A: Scatter plot of the causal relationship between “Never eat Sugar or foods/drinks containing sugar” and mania/bipolar disorder/manic depression. B: Scatter plot of the causal relationship between “Never eat Sugar or foods/drinks containing sugar” and depression. C: Scatter plot of the causal relationship between “Never eat Sugar or foods/drinks containing sugar” and schizophrenia. D: Leave-one-out plot of the causal relationship between “Never eat Sugar or foods/drinks containing sugar” and mania/bipolar disorder/manic depression. E: Leave-one-out plot of the causal relationship between “Never eat Sugar or foods/drinks containing sugar” and depression. F: Leave-one-out plot of the causal relationship between “Never eat Sugar or foods/drinks containing sugar” and schizophrenia.

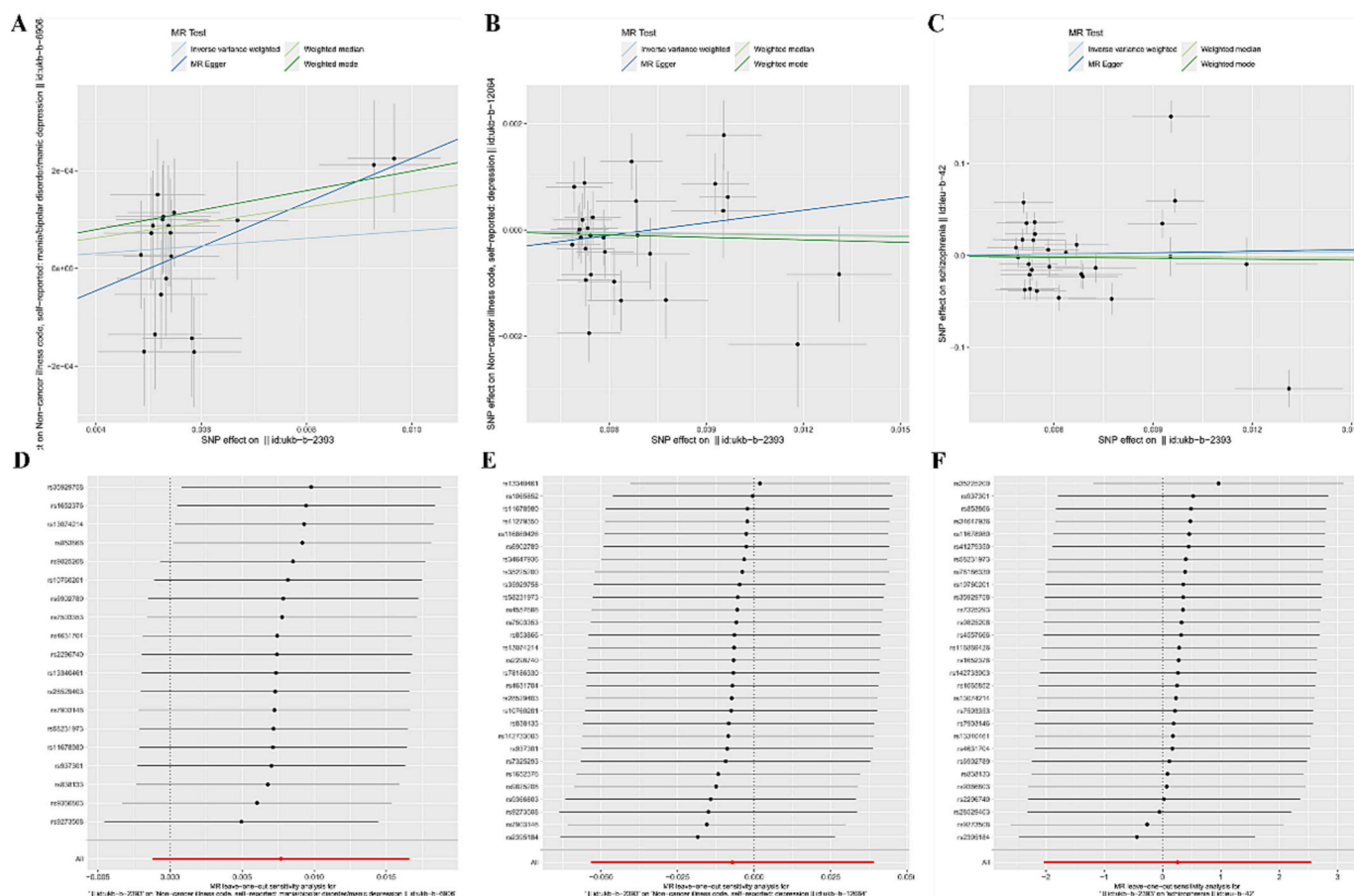
**Table 6**  
MR analysis of the causal relationship between Never eat eggs, dairy, wheat, sugar: I eat all of the above.

| Outcome phenotypes                      | MR method                 | Number of SNPs | Beta   | SE    | OR(95 % confidence Interval) | Association P-value | P for MR-Egger intercept |
|-----------------------------------------|---------------------------|----------------|--------|-------|------------------------------|---------------------|--------------------------|
| mania/bipolar disorder/manic depression | MR Egger                  | 19             | 0.045  | 0.021 | 1.046(1.005–1.089)           | 0.043               | 0.080                    |
|                                         | Weighted median           | 19             | 0.016  | 0.006 | 1.016(1.003–1.028)           | 0.013               |                          |
|                                         | Inverse variance weighted | 19             | 0.008  | 0.005 | 1.008(0.999–1.017)           | 0.091               |                          |
|                                         | Weighted mode             | 19             | 0.020  | 0.009 | 1.020(1.003–1.038)           | 0.034               |                          |
| depression                              | MR Egger                  | 29             | 0.078  | 0.090 | 1.081(0.906–1.290)           | 0.394               | 0.336                    |
|                                         | Weighted median           | 29             | −0.009 | 0.025 | 0.991(0.944–1.041)           | 0.730               |                          |
|                                         | Inverse variance weighted | 29             | −0.007 | 0.024 | 0.993(0.948–1.040)           | 0.762               |                          |
|                                         | Weighted mode             | 29             | −0.015 | 0.048 | 0.985(0.896–1.082)           | 0.752               |                          |
| schizophrenia                           | MR Egger                  | 29             | 0.557  | 4.693 | 1.745(0.0002–17,238.146)     | 0.907               | 0.948                    |
|                                         | Weighted median           | 29             | −0.115 | 0.761 | 0.892(0.201–3.966)           | 0.881               |                          |
|                                         | Inverse variance weighted | 29             | 0.258  | 1.171 | 1.294(0.130–12.846)          | 0.826               |                          |
|                                         | Weighted mode             | 29             | −0.317 | 1.920 | 0.728(0.017–31.375)          | 0.870               |                          |

wheat, sugar: I eat all of the above” and schizophrenia, 30 IVs related to “Never eat eggs, dairy, wheat, sugar: I eat all of the above” were initially selected from the GWAS database. After removing 1 abnormal IV with continuous imbalance, 29 IVs with genome-wide significant correlation were finally screened out. The horizontal pleiotropy was checked by MR Egger and MR-PRESSO, and it was believed that there was no horizontal pleiotropy ( $P$  for MR-Egger intercept = 0.948 > 0.05). The result of IVW

method showed that there was no causal relationship between “Never eat eggs, dairy, wheat, sugar: I eat all of the above” and schizophrenia ( $OR: 1.294, P > 0.05$ ). The other three methods further proved that there was no causal relationship between “Never eat eggs, dairy, wheat, sugar: I eat all of the above” and schizophrenia ( $P > 0.05$ ) (Table 6, Fig. 3C).

In the MR analysis of the causal relationship between “Never eat eggs, dairy, wheat, sugar: I eat all of the above” and the five mental



**Fig. 3.** Causal Relationship between “Never eat eggs, dairy, wheat, sugar: I eat all of the above” and Mental Disorder. A: Scatter plot of the causal relationship between “Never eat eggs, dairy, wheat, sugar: I eat all of the above” and mania/bipolar disorder/manic depression. B: Scatter plot of the causal relationship between “Never eat eggs, dairy, wheat, sugar: I eat all of the above” and depression. C: Scatter plot of the causal relationship between “Never eat eggs, dairy, wheat, sugar: I eat all of the above” and schizophrenia. D: Leave-one-out plot of the causal relationship between “Never eat eggs, dairy, wheat, sugar: I eat all of the above” and mania/bipolar disorder/manic depression. E: Leave-one-out plot of the causal relationship between “Never eat eggs, dairy, wheat, sugar: I eat all of the above” and depression. F: Leave-one-out plot of the causal relationship between “Never eat eggs, dairy, wheat, sugar: I eat all of the above” and schizophrenia.

disorders, all instrument variable F values were  $>10$ , indicating that all results did not have weak deviation, confirming that the results are reliable. The heterogeneity result  $P > 0.05$  indicates that there is no heterogeneity in the included IV, and the result does not need to consider the impact of heterogeneity. At the same time, all data were analyzed by “leave-one-out” analysis, and it was found that no single SNP had a large impact on the robustness of the results, so this study is stable (Fig. 3D, E, F). Therefore, the results of the MR analysis support that “Never eat eggs, dairy, wheat, sugar: I eat all of the above” has a causal relationship with mania/bipolar disorder/manic depression and there is a positive correlation ( $P < 0.05$ ); and there is no causal relationship with depression and schizophrenia.

#### 4. Discussion

This study adopted the Mendelian randomization (MR) method to systematically evaluate the causal relationship between diet and five major mental disorders. The results showed that ‘Never eat Wheat products’ has a causal relationship with the five mental disorders (mania, bipolar disorder, manic depression, depression, schizophrenia), all showing negative correlation. No significant causal relationship was observed between ‘Never eat Sugar or foods/drinks containing sugar’ and the five mental disorders. ‘Never eat eggs, dairy, wheat, sugar: I eat all of the above’ had a causal relationship with mania, bipolar disorder, and manic depression, showing a positive correlation; no significant causal relationship was observed with depression and schizophrenia.

Although some studies have reported possible links between diet and mental disorders, their results were mixed. Certain dietary patterns, like the Mediterranean diet, which is high in fruits, vegetables, legumes, fish, and unsaturated fats, have been linked to a lower risk of depression. On the other hand, diets high in processed foods and sugars, have been associated with a higher risk of depression and anxiety (Bremner et al., 2020; Michalak et al., 2012; Horn et al., 2022; Firth et al., 2020). While many studies suggest associations between diet and mental disorders, determining causality can be challenging. Many factors can confound these relationships, including socioeconomic status, physical health, and lifestyle habits like exercise and smoking. This is partly due to the inherent limitations of observational studies. Understanding whether an association is potentially causal or confounded is often difficult to achieve with traditional observational studies. The MR analysis method used in this study, which uses genetic variations to simulate randomization, is far less likely to be influenced by confounding factors, thus providing more reliable analytical results.

There was a relationship between wheat-based foods containing gluten and psychiatric diseases. Compared with participants on a diet containing gluten, those on a gluten-free diet showed improvements in both the Clinical Global Impression scale and negative symptoms, which suggested that the removal of gluten from the diet was associated with improvements in psychiatric and gastrointestinal symptoms in patients with schizophrenia or schizoaffective disorder (Kelly et al., 2019). Research has found that wheat products contain gluten, the main component of which is insoluble gliadin. This protein cannot be fully

broken down by the body, instead leaving behind small fragments of amino acids. These fragments can trigger allergic reactions in the body, damaging the villi of the small intestine, causing villous atrophy and inflammation, leading to celiac disease (Husby et al., 2020). Studies have reported a relationship between celiac disease and mental and psychological disorders, finding that people with celiac disease are more likely to have anxiety and depression than those without (Sharma et al., 2021). Compared to individuals without celiac disease, those with celiac disease have a significantly higher risk of developing schizophrenia, with the risk increasing by approximately twice (Wijarnprecha et al., 2018). Researchers completely eliminated foods containing gluten from the diets of psychiatric patients, which immediately resulted in significant symptom improvement (Clappison et al., 2020). The experiment also found that when gluten was reintroduced into the patient's diet, their recovery was impeded (Durazzo et al., 2022; Osman et al., 2021). It was also found that even without celiac disease, the body can still be allergic to gluten, which can impact the brain, causing it to function abnormally and lead to abnormal behavior (Martín-Masot et al., 2019).

This aligns with the finding in this study that 'Never eat Wheat products' can significantly reduce the risk of mania, bipolar disorder, manic depression, depression, schizophrenia. If gluten allergy is a cause of symptoms in some mental disorder patients, a simple test can be conducted and a gluten-free diet (GFD) provided to the patient. This may help alleviate symptoms, and maintaining a GFD is crucial for treating celiac disease and improving mental disorders. A GFD can also improve the efficacy of psychiatric medication treatments and can be used as a new treatment method, which may promote early diagnosis, treatment, and prevention of mental disorders.

Studies have shown that diets containing added sugars, soda, and junk food are associated with an increased risk of depression (Huang et al., 2019b; Lassale et al., 2019; Nicolaou et al., 2020). However, this study did not observe a significant causal relationship between 'Never eat Sugar or foods/drinks containing sugar' and mental disorders. Clinically, most mental disorder patients are observed to have gastrointestinal allergies caused by eating eggs, dairy, wheat, etc., which can easily lead to a leaky gut, an important factor in the onset of mental disorders. This study found that 'Never eat eggs, dairy, wheat, sugar: I eat all of the above' has a causal relationship with mania, bipolar disorder, manic depression, showing a positive correlation, suggesting that eating eggs, dairy, wheat, sugar might increase the risk of mania, bipolar disorder, manic depression. It is recommended that patients with mania, bipolar disorder, manic depression control their intake of eggs, dairy, wheat, sugar as much as possible; no significant causal relationship was observed with depression and schizophrenia.

There were still some limitations in this study. Firstly, in Two-Sample Mendelian Randomization (TSMR) analyses, the exposure and outcome studies generally do not involve overlap. However, this study might be influenced by some overlap between exposure and outcome studies. To mitigate this bias, an F-test (with values >10) was employed to minimize sample overlap. Secondly, the GWAS data utilized in this study originated from European populations, thereby constraining the generalizability of these findings to other ethnic groups. Further research validation is needed to ascertain the applicability of the results to diverse populations. Thirdly, the TSMR analysis did not incorporate gender stratification, thus precluding an exploration of potential gender differences. It is recommended that GWAS databases incorporate gender-specific data for subsequent research investigations.

## 5. Conclusion

In summary, this study uses the Mendelian randomization method to systematically analyze the causal relationship between diet and five major mental disorders. It was found that 'Never eat Wheat products' has a negative correlation with these five mental disorders (mania, bipolar disorder, manic depression, depression, schizophrenia), suggesting that avoiding wheat products may reduce the risk of mental disorder.

It is suggested that clinicians could provide a gluten-free diet (GFD) to patients with mental disorders, which might help alleviate symptoms. Maintaining a GFD is crucial for treating celiac disease and improving mental disorders, and GFD can also improve the efficiency of psychiatric medication treatments. It can be used as a new treatment method, potentially promoting early diagnosis, treatment, and prevention of mental disorders.

Additionally, 'Never eat eggs, dairy, wheat, sugar: I eat all of the above' has a positive correlation with mania, bipolar disorder, manic depression, indicating that eating eggs, dairy, wheat, sugar might increase the risk of mania, bipolar disorder, manic depression. There was no significant causal relationship observed with depression and schizophrenia. It is recommended that patients with mania, bipolar disorder, and manic depression control their intake of eggs, dairy, wheat, sugar as much as possible. Furthermore, due to genetic differences among different ethnicities, countries, and regions, further research is needed for different populations. If subsequent related GWAS studies could find more new dietary phenotypes and genetic variations related to mental disorders, MR studies can use more SNPs while increasing the sample size to obtain more reliable conclusions.

## Ethics approval and consent to participate

Not applicable.

## Consent for publication

Not applicable.

## CRediT authorship contribution statement

Zhiqiang Du, Zhenhe Zhou and Haohao Zhu conceived the study; Shuaiyi Guo, Yifan Sun, Ying Jiang, Qin Zhou and Hongliang Zhou collected the report; Zhiqiang Du, Shuaiyi Guo, Yifan Sun and Haohao Zhu wrote the manuscript and edited the manuscript; Zhenhe Zhou and Hongliang Zhou revised the manuscript. All authors have approved publication of the manuscript.

## Funding

The work is supported by the National Natural Science Foundation of China (No. 82104244), Wuxi Municipal Health Commission (No. Q202101 and ZH202110), Wuxi Taihu Talent Project (No. WXTTP2020008 and WXTTP2021), Wuxi Medical Development Discipline Project (No. FZXK2021012) and Jiangsu Research Hospital Association for Precision Medication (JY202105).

## Declaration of competing interest

The authors declare no conflicts of interest.

## Data availability

The dataset generated during and analyzed during the current study are available from the MR Base database (<http://www.mrbase.org/>).

## Acknowledgments

We thank Brian Springer (University of New England, Email: [bspringer1@une.edu](mailto:bspringer1@une.edu)) for the linguistic editing and proofreading of the manuscript.

## References

- Aucoin, M., LaChance, L., Cooley, K., et al., 2020. Diet and psychosis: a scoping review [J]. *Neuropsychobiology* 79 (1), 20–42.



- Bly, M.J., Taylor, S.F., Dalack, G., et al., 2014. Metabolic syndrome in bipolar disorder and schizophrenia: dietary and lifestyle factors compared to the general population [J]. *Bipolar Disord.* 16 (3), 277–288.
- Bremner, J.D., Moazzami, K., Wittbrodt, M.T., et al., 2020. Diet, stress and mental health [J]. *Nutrients* 12 (8), 2428.
- Clappison, E., Hadjivassiliou, M., Zis, P., 2020. Psychiatric manifestations of coeliac disease, a systematic review and meta-analysis[J]. *Nutrients* 12 (1), 142.
- Cooper, C., 2019. Global, regional, and national burden of neurological disorders, 1990–2016: a systematic analysis for the Global Burden of Disease Study 2016[J]. *Lancet Neurol.* 18 (4), 357–375.
- Dickerson, F., Gennusa III, J.V., Stallings, C., et al., 2020. Protein intake is associated with cognitive functioning in individuals with psychiatric disorders[J]. *Psychiatry Res.* 284, 112700.
- Durazzo, M., Ferro, A., Brascugli, I., et al., 2022. Extra-intestinal manifestations of celiac disease: what should we know in 2022?[J]. *J. Clin. Med.* 11 (1), 258.
- Firth, J., Stubbs, B., Teasdale, S.B., et al., 2018. Diet as a hot topic in psychiatry: a population-scale study of nutritional intake and inflammatory potential in severe mental illness[J]. *World Psychiatry* 17 (3), 365.
- Firth, J., Solmi, M., Wootton, R.E., et al., 2020. A meta-review of “lifestyle psychiatry”: the role of exercise, smoking, diet and sleep in the prevention and treatment of mental disorders[J]. *World Psychiatry* 19 (3), 360–380.
- Gili, M., Castellví, P., Vives, M., et al., 2019. Mental disorders as risk factors for suicidal behavior in young people: a meta-analysis and systematic review of longitudinal studies[J]. *J. Affect. Disord.* 245, 152–162.
- Hemani, G., Zheng, J., Wade, K.H., et al., 2016. MR-Base: a platform for systematic causal inference across the phenome using billions of genetic associations. *eLife*, doi: 10.7554/eLife.34408.
- Hemani, G., Zheng, J., Elsworth, B., et al., 2018. The MR-Base platform supports systematic causal inference across the human phenome[J]. *eLife* 7, e34408.
- Hepsomali, P., Coxon, C., 2022. Inflammation and diet: focus on mental and cognitive health[J]. *Adv. Clin. Exp. Med.* 31 (8), 821–825.
- Horn, J., Mayer, D.E., Chen, S., et al., 2022. Role of diet and its effects on the gut microbiome in the pathophysiology of mental disorders[J]. *Transl. Psychiatry* 12 (1), 164.
- Huang, Y., Wang, Y.U., Wang, H., et al., 2019a. Prevalence of mental disorders in China: a cross-sectional epidemiological study[J]. *Lancet Psychiatry* 6 (3), 211–224.
- Huang, Q., Liu, H., Suzuki, K., et al., 2019b. Linking what we eat to our mood: a review of diet, dietary antioxidants, and depression[J]. *Antioxidants* 8 (9), 376.
- Husby, S., Koletzko, S., Korponay-Szabó, I., et al., 2020. European society paediatric gastroenterology, hepatology and nutrition guidelines for diagnosing coeliac disease 2020[J]. *J. Pediatr. Gastroenterol. Nutr.* 70 (1), 141–156.
- Karyotaki, E., Cuijpers, P., Albor, Y., et al., 2020. Sources of stress and their associations with mental disorders among college students: results of the world health organization world mental health surveys international college student initiative[J]. *Front. Psychol.* 11, 1759.
- Kelly, D.L., Demyanovich, H.K., Rodriguez, K.M., et al., 2019. Randomized controlled trial of a gluten-free diet in patients with schizophrenia positive for antigliadin antibodies (AGA IgG): a pilot feasibility study[J]. *J. Psychiatry Neurosci.* 44 (4), 269–276.
- Lassale, C., Batty, G.D., Baghdadli, A., et al., 2019. Healthy dietary indices and risk of depressive outcomes: a systematic review and meta-analysis of observational studies [J]. *Mol. Psychiatry* 24 (7), 965–986.
- Liu, N.H., Daumit, G.L., Dua, T., et al., 2017. Excess mortality in persons with severe mental disorders: a multilevel intervention framework and priorities for clinical practice, policy and research agendas[J]. *World Psychiatry* 16 (1), 30–40.
- Ljungberg, T., Bondza, E., Lethin, C., 2020. Evidence of the importance of dietary habits regarding depressive symptoms and depression[J]. *Int. J. Environ. Res. Public Health* 17 (5), 1616.
- Lv, W., He, J., Shao, J., et al., 2023. Causal relationships between short-chain fatty acids and L-isoleucine biosynthesis and susceptibility and severity of COVID-19: evidence from Mendelian randomization[J]. *J. Infect.* S0163-4453 (23), 00258.
- Martín-Masot, R., Nestares, M.T., Diaz-Castro, J., et al., 2019. Multifactorial etiology of anemia in celiac disease and effect of gluten-free diet: a comprehensive review[J]. *Nutrients* 11 (11), 2557.
- Michalak, J., Zhang, X.C., Jacobi, F., 2012. Vegetarian diet and mental disorders: results from a representative community survey[J]. *Int. J. Behav. Nutr. Phys. Act.* 9 (1), 1–10.
- Nicolaou, M., Colpo, M., Vermeulen, E., et al., 2020. Association of a priori dietary patterns with depressive symptoms: a harmonised meta-analysis of observational studies[J]. *Psychol. Med.* 50 (11), 1872–1883.
- Nightingale, B.A., Cassin, S.E., 2019. Disordered eating among individuals with excess weight: a review of recent research[J]. *Curr. Obes. Rep.* 8, 112–127.
- Osman, D., Umar, S., Muhammad, H., et al., 2021. Neurological manifestation of coeliac disease with particular emphasis on gluten ataxia and immunological injury: a review article[J]. *Gastroenterol. Hepatol. Bed Bench* 14 (1), 1.
- Sagar, R., Dandona, R., Gururaj, G., et al., 2020. The burden of mental disorders across the states of India: the Global Burden of Disease Study 1990–2017[J]. *Lancet Psychiatry* 7 (2), 148–161.
- Sharma, N., Singh, K., Senapati, S., 2021. Celiac disease poses significant risk in developing depression, anxiety, headache, epilepsy, panic disorder, dysthymia: a meta-analysis[J]. *Indian J. Gastroenterol.* 1–10.
- Teasdale, S.B., Ward, P.B., Samaras, K., et al., 2019. Dietary intake of people with severe mental illness: systematic review and meta-analysis[J]. *Br. J. Psychiatry* 214 (5), 251–259.
- Verbanck, M., Chen, C.Y., Neale, B., et al., 2018. Detection of widespread horizontal pleiotropy in causal relationships inferred from Mendelian randomization between complex traits and diseases[J]. *Nat. Genet.* 50 (5), 693–698.
- Wijarnpreecha, K., Jaruvongvanich, V., Cheungpasitporn, W., et al., 2018. Association between celiac disease and schizophrenia: a meta-analysis[J]. *Eur. J. Gastroenterol. Hepatol.* 30 (4), 442–446.
- Woolf, B., Di Cara, N., Moreno-Stokoe, C., et al., 2022. Investigating the transparency of reporting in two-sample summary data Mendelian randomization studies using the MR-Base platform[J]. *Int. J. Epidemiol.* 51 (6), 1943–1956.
- Wu, A., March, L., Zheng, X., et al., 2020. Global low back pain prevalence and years lived with disability from 1990 to 2017: estimates from the Global Burden of Disease Study 2017[J]. *Ann. Transl. Med.* 8(6).